

Soluções - FME, Volume 06

Capítulo I - Exercícios

1. a) $(6 + 7i)(1 + i) = (6, 7)(1, 1) = (6 - 7, 6 + 7) = -1 + 13i$
b) $(5 + 4i)(1 - i) + (2 + i)i = (5, 4)(1, -1) + (2, 1)(0, 1) = (5 + 4 + 0 - 1, -5 + 4 + 2 - 0) = 8 + i$
c) $(1 + 2i)^2 - (3 + 4i) = (1, 2)^2 - (3, 4) = (1 - 2^2, 4) - (3, 4) = -6$
2. a) $(3 + 2i) + (2 - 5i) = 5 - 3i$
b) $(5 - 2i) - (2 + 8i) = 3 - 10i$
c) $(1 + i) + (1 - i) - 2i = 2 - 2i$
d) $(6 + 7i) - (4 + 2i) + (1 - 10i) = 3 - 5i$
3. a) $(2 - 3i)(1 + 5i) = (2 + 15, 10 - 3) = 17 + 7i$
b) $(1 + 2i)(2 + i) = (2 - 2, 1 + 4) = 5i$
c) $(4 - 3i)(5 - i)(1 + i) = (20 - 3, -4 - 15)(1, 1) = (17 + 19, 17 - 19) = 36 - 2i$
d) $(7 + 2i)(7 - 2i) = (49 + 4, -14 + 14) = 53$
4. $u = 4 + 3i, v = 5 - 2i, uv = ?$
 $uv = (4, 3)(5, -2) = (20 + 6, -8 + 15) = 26 + 7i$
5. a) $(3 + 2i)^2 = (3, 2)^2 = (3^2 - 2^2, 2 \cdot 6) = 5 + 12i$
b) $(5 - i)^2 = (5, -1)^2 = (5^2 - (-1)^2, -10) = 24 - 10i$
c) $(1 + i)^3 = (1, 1)^3 = (1^2 - 1^2, 2)(1, 1) = (0 - 2, 0 + 2) = -2 + 2i$
6. $f(z) = z^2 - z + 1, f(1 - i) = ?$
 $f(1 - i) = (1, -1)^2 - (1, -1) + (1, 0) = (0, -2) + (0, 1) = -i$
7. $f(z) = z^4 + iz^3 - (1 + 2i)z^2 + 3z + 1 + 3i, f(1 + i) = ?$
 $f(1 + i) = (1, 1)^2 = (0, 2)$
 $= (0, 2)^2 + (0, 1)(0, 2)(1, 1) - (1, 2)(0, 2) + (3, 3) + (1, 3)$
 $= (-4, 0) + (-2, 0)(1, 1) - (-4, 2) + (4, 6)$
 $= 2[(-2, 0) + (-1, -1) - (-2, 1) + (2, 3)]$

$$= 2(1, 1) = 2(1 + i)$$

8. Prove que $(1 + i)^2 = 2i$ e coloque na forma algébrica $z = \frac{(1 + i)^{80} - (1 + i)^{82}}{i^{96}}$

$$1. (1, 1)^2 = (1 - 1, 1 + 1) = 2i$$

$$2. z = \frac{(2i)^{40} - (2i)^{41}}{(i^2)^{48}} = \frac{2^{40} - 2^{41} \cdot i}{1} = 2^{40} - 2^{41}i$$

$$9. a) i^{76} = (i^2)^{38} = 1$$

$$b) i^{110} = (i^2)^{55} = -1$$

$$c) i^{97} = (i^2)^{48} \cdot i = i$$

$$d) i^{503} = (i^2)^{251} \cdot i = -i$$

10. Prove que $(1 - i)^2 = -2i$ e calcule $(1 - i)^{96} + (1 - i)^{97}$

$$(1, -1)^2 = (1 - (-1)^2, -1 - 1) = -2i$$

$$(1, -1)^{96} + (1, -1)^{97} = (-2i)^{48} + (-2i)^{48}(1, -1) = 2^{48} + 2^{48} - 2^{48}i = 2^{49} - 2^{48}i$$

11. Se $i^2 = -1$, calcule $(1 + i)^{12} - (1 - i)^{12}$

$$= (2i)^6 - (-2i)^6 = -2^6 + 2^6 = 0$$

12. Simplifique $\frac{(2 + i)^{101}(2 - i)^{50}}{(-2 - i)^{100}(i - 2)^{49}}$

$$= \frac{(2, 1)^{101}(2, -1)^{50}}{(-2, 1)^{100}(-2, -1)^{49}} = \frac{(2, 1)^{101}(2, -1)^{50}}{(2, 1)^{100}(-1)(2, -1)^{49}} = -(2, 1)(2, -1) = -(4 + 1, -2 + 2) = -5$$

13. A igualdade $(1 + i)^n = (1 - i)^n$ é válida para os números naturais divisíveis por qual natural?

$$\bullet \text{ Para } n = 2k, k \in \mathbb{N} \Rightarrow ((1 + i)^2)^k = ((1 - i)^2)^k \Rightarrow (2i)^k = (-2i)^k \Rightarrow 2^k i^k = (-1)^k 2^k i^k$$

$$\Rightarrow (-1)^k = 1 \Rightarrow k = 2k'$$

$$\Rightarrow n = 4k'', k'' \in \mathbb{N}$$

$$\bullet \text{ Para } n = 2k + 1, k \in \mathbb{N} \Rightarrow ((1 + i)^2)^k(1 + i) = ((1 - i)^2)^k(1 - i) \Rightarrow (2i)^k(1 + i) = (-2i)^k(1 - i)$$

$$\Rightarrow \frac{(1 + i)(1 + i)}{(1 - i)(1 + i)} = (-1)^k \Rightarrow (-1)^k = i \Rightarrow \perp$$

$\therefore n$ deve ser divisível por 4.

$$14. a) 2 + 3yi = x + 9i \quad x = 2, 3y = 9 \Rightarrow \begin{cases} x = 2 \\ y = 3 \end{cases}$$

$$\text{b) } (x + yi)(3 + 4i) = 7 + 26i \quad (3x - 4y, 4x + 3y) = (7, 26) \Rightarrow \begin{cases} 3x - 4y = 7 \\ 4x + 3y = 26 \end{cases}$$

$$\cancel{3x} - 4y - \frac{3}{4}\cancel{4x} - \frac{3}{4} \cdot 3y = 7 - \frac{3}{4} \cdot 26 \Rightarrow -4y - \frac{9}{4}y = 7 - \frac{3 \cdot 26}{4} \Rightarrow -y \left(\frac{4^2 + 3^2}{4} \right) = \frac{28 - 78}{4}$$

$$y = \frac{50}{25} = 2$$

$$\Rightarrow x = \frac{7 + 4 \cdot 2}{3} = \frac{15}{3} = 5 \Rightarrow \begin{cases} x = 5 \\ y = 2 \end{cases}$$

$$\text{c) } (x + yi)^2 = 4i \quad (x^2 - y^2, 2xy) = (0, 4) \Rightarrow \begin{cases} x^2 = y^2 \Rightarrow x = y \\ xy = 2 \end{cases} \Rightarrow x^2 = 2 \Rightarrow \begin{cases} x = y = \sqrt{2} \\ \text{ou} \\ x = y = -\sqrt{2} \end{cases}$$

$$15. \text{ a) } 3 + 5ix = y - 15i \Rightarrow \begin{cases} y = 3 \\ 5x = -15 \end{cases} \Rightarrow \begin{cases} x = -3 \\ y = 3 \end{cases}$$

$$\text{b) } (x + yi)(2 + 3i) = 1 + 8i \quad (2x - 3y, 3x + 2y) = (1, 8) \Rightarrow \begin{cases} 2x - 3y = 1 \\ 3x + 2y = 8 \end{cases}$$

$$\cancel{2x} - 3y - \frac{2}{3}\cancel{3x} - \frac{2}{3} \cdot 2y = 1 - \frac{2}{3} \cdot 8 \Rightarrow y \left(\frac{-9 - 4}{3} \right) = \frac{3 - 16}{3} \Rightarrow y = \frac{-13}{-13} = 1$$

$$\Rightarrow x = \frac{1 + 3}{2} \Rightarrow \begin{cases} x = 2 \\ y = 1 \end{cases}$$

$$\text{c) } (3 + yi) + (x - 2i) = 7 - 5i \Rightarrow \begin{cases} 3 + x = 7 \\ y - 2 = -5 \end{cases} \Rightarrow \begin{cases} x = 4 \\ y = -3 \end{cases}$$

$$\text{d) } (x + yi)^2 = 2i \Rightarrow (x^2 - y^2, 2xy) = (0, 2) \Rightarrow \begin{cases} x^2 = y^2 \\ xy = 1 \end{cases} \Rightarrow \begin{cases} x = 1 \text{ e } y = 1 \\ \text{ou} \\ x = -1 \text{ e } y = -1 \end{cases}$$

$$\text{e) } (2 - x + 3y) + 2yi = 0 \Rightarrow \begin{cases} 2 - x + 3y = 0 \\ 2y = 0 \end{cases} \Rightarrow \begin{cases} x = 2 \\ y = 0 \end{cases}$$

$$\text{f) } (3 - i)(x + yi) = 20 \quad (3x + y, 3y - x) = (20, 0) \Rightarrow \begin{cases} 3x + y = 20 \\ 3y = x \end{cases}$$

$$\Rightarrow 3x + \frac{x}{3} = 20 \Rightarrow 9x + x = 3 \cdot 20 \Rightarrow x = 6 \Rightarrow \begin{cases} x = 6 \\ y = 2 \end{cases}$$

$$16. \begin{cases} x + yi = i \\ xi + y = 2i - 1 \end{cases}$$

$$x = i(1 - y) \Rightarrow (1 - y)i^2 + y = 2i - 1 \Rightarrow y - 1 + y = 2i - 1 \Rightarrow y = i$$

$$\Rightarrow x + i^2 = i \Rightarrow x = 1 + i \Rightarrow \begin{cases} x = 1 + i \\ y = i \end{cases}$$

17. $(a, b)(c, d) = (ac - bd, ad + bc) = (x, 0), x \in \mathbb{R}$

$$\Rightarrow ad + bc = 0 \Rightarrow ad = -bc \Rightarrow \frac{a}{c} = -\frac{b}{d}$$

18. $(a, b)^2 \Rightarrow (a^2 - b^2, 2ab)^2 = ((a^2 - b^2)^2 - 4a^2b^2, 4ab(a^2 - b^2))$

$$\Rightarrow 1. 4ab(a^2 - b^2) = 0 \Rightarrow a = \pm b \ (a \neq 0 \wedge b \neq 0) \text{ ou } a = 0 \text{ ou } b = 0$$

$$\Rightarrow 2. (a^2 - b^2)^2 - 4a^2b^2 < 0 \Rightarrow (a^2 - b^2)^2 < 4a^2b^2$$

- $a = 0, b \neq 0 \Rightarrow (-b^2)^2 < 0 \Rightarrow b^4 < 0 \Rightarrow \perp$

- $a \neq 0, b = 0 \Rightarrow a^4 < 0 \Rightarrow \perp$

- $a = \pm b, a \neq 0, b \neq 0 \Rightarrow ((\pm b)^2 - b^2)^2 < 4(\pm b)^2b^2 \Rightarrow 0 < 4b^4, \forall b \neq 0$

$$\therefore \begin{cases} a = \pm b \\ ab \neq 0 \end{cases}$$

19. a) $\frac{2}{i} \cdot \frac{-i}{-i} = -2i$

b) $\frac{3}{2+i} \cdot \frac{2-i}{2-i} = \frac{6}{5} - \frac{3}{5}i$

c) $\frac{1+2i}{3-i} \cdot \frac{3+i}{3+i} = \frac{1}{10} + \frac{7}{10}i$

d) $\frac{i^9}{4-3i} \cdot \frac{4+3i}{4+3i} = \frac{i^8 \cdot i(4+3i)}{25} = \frac{-3}{25} + \frac{4}{25}i$

20. a) $\frac{1}{i} \cdot \frac{i}{i} = -i$

b) $\frac{1}{1+i} \cdot \frac{1-i}{1-i} = \frac{1}{2} - \frac{1}{2}i$

c) $\frac{3+4i}{2-i} \cdot \frac{2+i}{2+i} = \frac{2}{3} + \frac{11}{5}i$

d) $\frac{1+i}{1-i} \cdot \frac{1+i}{1+i} = \frac{2i}{2} = i$

e) $\frac{i^{11} + 2i^{13}}{i^{18} - i^{37}} = \frac{i^{11}(1 + 2i^2)}{i^{18}(1 - i^{19})} = \frac{-1}{i^4 i^2 i(1 - i^{16} i^2 i)} = \frac{1}{i(1+i)} = \frac{1}{-1+i} \cdot \frac{-1+i}{-1+i} = -\frac{1}{2} - \frac{1}{2}i$

f) $\frac{1-3i}{3-i} \cdot \frac{3+i}{3+i} = \frac{3}{5} - \frac{4}{5}i$

g) $\frac{i^3 - i^2 + i^{17} - i^{35}}{i^{16} - i^{13} + i^{30}} = \frac{i^2(i - 1 + i^{12}i^3 - i^{32}i)}{i^{13}(i^3 - 1 + i^{17})} = \frac{i - 1 - i - i}{i^3(-i - 1 + i^{16}i)} = \frac{-(1+i)}{-i(-1)} \cdot \frac{-i}{-i} = -1 + i$

h) $\frac{1+i}{(1-i)^2} = \frac{1+i}{-2i} \cdot \frac{2i}{2i} = \frac{-2+2i}{4} = -\frac{1}{2} + \frac{1}{2}i$

$$21. \frac{1+3i}{2-i} \cdot \frac{2+i}{2+i} = -\frac{1}{5} - \frac{7}{5}i$$

$$22. \frac{1+i}{i} \cdot \frac{-i}{-i} = 1-i \implies 1+i$$

$$23. z = \frac{1-i}{1+i} \cdot \frac{-2i}{2} = -i \implies z^{-1} = \frac{1}{-i} \cdot \frac{i}{i} = i \implies \overline{z^{-1}} = -i$$

$$24. u^2 - v^2 = 6, \bar{u} + \bar{v} = 1-i, u-v=?$$

$$(u-v)(u+v) = 6 \implies u-v = \frac{6}{u+v}$$

$$\bar{u} + \bar{v} = 1-i \implies \overline{u+v} = \overline{1-i} \implies u+v = 1+i$$

$$\therefore u-v = \frac{6}{u+v} = \frac{6}{1+i} \cdot \frac{1-i}{1-i} = 3-3i$$

$$25. \text{Provavelmente tem uma informação errada na questão } u = 1$$

$$26. \left(\frac{1+i}{1-i} \right)^n, n \in \mathbb{Z}$$

$$\left(\frac{1+i}{1-i} \cdot \frac{1+i}{1+i} \right)^n = \left(\frac{2i}{2} \right)^n = i^n \therefore 1, i, -1, -i$$

$$27. \frac{a+bi}{c+di}, c+di \neq 0$$

$$\frac{a+bi}{c+di} \cdot \frac{c-di}{c-di} = \frac{ac-bd+i(ad+bc)}{c^2+d^2}$$

$$\implies ax-bd=0$$

$$\implies ad+bc=0$$

$$28. u = x + yi, v = \frac{1}{2} - \frac{\sqrt{3}}{2}i, \text{Re}(v\bar{u}) = ?$$

$$v\bar{u} = (x+yi)\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = \left(\frac{x}{2} - \frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x + \frac{y}{2}\right)$$

$$\text{Re}(v\bar{u}) = \frac{x}{2} - \frac{\sqrt{3}}{2}y$$

$$29. \dots$$

$$30. z_1, z_2 \in \mathbb{C}, \text{Im}(z_1 + z_2) = \text{Im}(z_1 z_2) = 0$$

$$z_1 = a+bi, z_2 = c+di \implies \begin{cases} z_1 z_2 = (ac-bd, ad+bc) \\ z_1 + z_2 = (a+c, b+d) \end{cases} \implies \begin{cases} ad+bc=0 \\ b+d=0 \end{cases}$$

$$\implies \begin{cases} ad = -bc \\ b = -d \end{cases} \implies \begin{cases} a = c \\ b = -d \end{cases} \implies z_1 = \bar{z}_2 \text{ ou } z_1, z_2 \in \mathbb{R}$$

31. $z = \frac{2 - xi}{1 + 2xi}, x \in \mathbb{R}, \operatorname{Re}(z) = 0$

$$z = \frac{2 - xi}{1 + 2xi} \cdot \frac{1 - 2xi}{1 - 2xi} = \frac{(2x - 2x^2) + (-4x - x)i}{1 + 4x^2}$$

$$z = \frac{2(1 - x^2) - 3xi}{4x^2 + 1} \implies \frac{2(1 - x^2)}{4x^2 + 1} = 0 \implies 1 - x^2 = 0 \implies x = \pm 1$$

32. $z = \frac{1 + 2i}{2 + ai}, a = ?$

$$\frac{1 + 2i}{2 + ai} \cdot \frac{2 - ai}{2 - ai} = \frac{(2 + 2a) + (-a + 4)i}{a^2 + 4} \implies \frac{-a + 4}{a^2 + 4} = 0 \implies a = 4$$

33. $\operatorname{Im}(z(5 + 8i)) = 0, \operatorname{Re}\left(\frac{z}{1 + i}\right) = 0, z = ?$

$$(x + yi)(5 + 8i) = (5x - 8y) + (8x + 5y)i \implies 8x + 5y = 0$$

$$\frac{x + yi}{1 + i} \cdot \frac{1 - i}{1 - i} = \frac{(x + y) + (-x + y)i}{2} \implies \frac{x + y}{2} = 0 \implies x + y = 0$$

$$\implies \perp$$

34. ...

35. $z \in \mathbb{C}, \bar{z} = -2zi$

$$z = x + yi \implies x - yi = -2i(x + yi) = -2xi + 2y \implies \begin{cases} x = 2y \\ y = 2x \end{cases} \implies z = 0$$

36. $z = x + yi, u = \frac{1}{2} - \frac{\sqrt{3}}{2}i, \operatorname{Re}(u\bar{z}) = ?, \operatorname{Im}(u\bar{z}) = ?$

$$u\bar{z} = \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)(x - yi) = \left(\frac{x}{2} - \frac{\sqrt{3}}{2}y, \frac{-y}{2} - \frac{\sqrt{3}}{2}x\right)$$

$$\begin{cases} \operatorname{Re}(u\bar{z}) = \frac{x}{2} - \frac{\sqrt{3}}{2}y \\ \operatorname{Im}(u\bar{z}) = \frac{-\sqrt{3}}{2}x - \frac{y}{2} \end{cases}$$

37. Prove que $\overline{z^n} = (\bar{z})^n, \forall n \in \mathbb{N}$.

$$z = a + bi \implies z^n = (a + bi)^n = \sum_k^n \binom{n}{k} a^k (bi)^{n-k}$$

$$\implies \overline{z^n} = \sum_k^n \binom{n}{k} a^k (-bi)^{n-k} = (a + (-bi))^n = (\bar{z})^n$$

38. ...

39. $z\bar{z} + (z - \bar{z}) = 13 + 6i, z = ?$

$$x^2 + y^2 + \cancel{x} + yi - \cancel{x} + yi = 13 + 6i \implies x^2 + y^2 + 2yi = 13 + 6i \implies \begin{cases} x^2 + y^2 = 13 \\ 2y = 6 \end{cases}$$

$$\implies x^2 = 13 - 9 = 4 \implies \begin{cases} x = \pm 2 \\ y = 3 \end{cases}$$

40. ...

41. ...